

Project Documentation

1. Introduction

- **Project Title:** Auto Ticket Classification using Flow Designer in ServiceNow
- **Team Members:**
 - Lead Developer / Flow Designer Specialist – *Configuration, Flow Logic, and Backend Setup*
 - ServiceNow Administrator – *Table Structure, Update Sets, and Deployment*

2. Project Overview

- **Purpose:** IT staff at a school helpdesk manually review and assign categories to daily support requests (e.g., Wi-Fi, projector failures, password resets, slow computers), making the process time-consuming and prone to human error. The purpose of this project is to automate ticket categorization at creation to reduce manual effort and accelerate response times.
- **Features:**
 - Automated classification of incoming tickets using keyword rules in Flow Designer.
 - Custom dynamic choice mapping for categories and subcategories.
 - Automatic email notification sent to the requester upon ticket creation.
 - Centralized update set management for seamless deployment.

3. Architecture

- **Frontend:** ServiceNow Service Portal / Workspace interface allowing callers to submit and view incident records.
- **Backend:** ServiceNow Flow Designer engine executing server-side logic based on trigger conditions without custom code.
- **Database:** ServiceNow relational database hosting custom table `u_incident_workflow` referencing system user (`sys_user`) and user group (`sys_user_group`) tables.

4. Setup Instructions

- **Prerequisites:**
 - ServiceNow Instance (Utah/Washington/Xanadu or later) with Admin Access.
 - Flow Designer plugin enabled.
- **Installation:**
 1. Navigate to **All menus** → **System Update Sets** → **Local Update Sets**.
 2. Create a new Update Set named **Project Update Set** (State: *In progress*, Application: *Global*).
 3. Click **Make This My Current Set** under Related Links.

5. Folder Structure / Data Model Structure

Table: Incident WorkFlow (u_incident_workflow)

Field Name	Column Name	Data Type	Reference / Configuration
Number	u_number	Auto Number	Prefix: INC, Number: 500, Digits: 5
Caller	u_caller	Reference	sys_user
Category	u_category	Choice	Network, Hardware, Access, Performance
Subcategory	u_subcategory	Choice	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	u_short_description	String	Max length 160
Description	u_description	String	String (Full text)
State	u_state	Choice	New, In progress, On hold, Resolved, Closed

Field Name	Column Name	Data Type	Reference / Configuration
Assigned Group	u_assigned_group	Reference	sys_user_group
Assigned To	u_assigned_to	Reference	sys_user

Dependent Choice Logic Configuration

- **Dependent Field:** u_subcategory depends on u_category.
- **Mappings:**
 - Wi-Fi → network
 - Projector → hardware
 - Forgot Password → access
 - Slow Computer → performance

6. Running the Application (Flow Execution)

1. Open **Flow Designer** in ServiceNow.
2. Open Flow: Auto Classify School IT Tickets.
3. Click **Activate** to make the flow active in the current instance.
4. Submit a record to u_incident_workflow with Category left empty.

7. API / Integration & Flow Logic Documentation

Trigger

- **Table:** Incident WorkFlow (u_incident_workflow)
- **Condition:** Created where Category is empty

Rules Engine Matrix

Keyword Criteria (Short Description contains)	Output Category	Output Subcategory
Wi-Fi OR Network	Network	Wi-Fi
Projector OR Hardware	Hardware	Projector
Forgot password	Access	Forgot Password

Keyword Criteria (Short Description contains)	Output Category	Output Subcategory
Slow Computer	Performance	Slow Computer

Action Engine

- **Update Record:** Assign mapped Category and Subcategory values to the triggered record.
- **Send Email:**
 - **To:** Trigger Record → Caller → Email
 - **Subject:** Your Request for the issue has been submitted.
 - **Body:** Automated email acknowledgment confirming receipt and classification.

8. Authentication & Security

- Access to `u_incident_workflow` is controlled via standard ServiceNow **Access Control Lists (ACLs)**.
- Requesters must be authenticated standard users logged into the ServiceNow instance.
- Flow Designer executes under system context with security permissions managed through ServiceNow platform roles (`itil`, `admin`).

9. User Interface

- **Caller Form View:** Simple entry form containing *Caller*, *Short Description*, and *Description*.
- **Fulfillment View:** Complete record view displaying populated *Category*, *Subcategory*, *Assigned Group*, and *State*.

10. Testing & Validation

Test Case	Input (Short Description)	Expected Output	Status
TC-01	"Wi-Fi not working in library"	Category: Network , Subcategory: Wi-Fi , Email sent	Pass
TC-02	"Projector not turning on"	Category: Hardware , Subcategory: Projector , Email sent	Pass
TC-03	"Forgot password for portal"	Category: Access , Subcategory: Forgot Password , Email sent	Pass
TC-04	"Slow Computer in lab 2"	Category: Performance , Subcategory: Slow Computer , Email sent	Pass

11. Demonstration & Export Instructions

1. Navigate to **System Update Sets** → **Local Update Sets** and open Project Update Set.
2. Set **State** to **Complete** and click **Save**.
3. Select **Export to XML** from Related Links to download the update package for migration across instances (Dev → Test → Prod).

12. Known Issues

- Tickets submitted with non-matching short descriptions (e.g., *"Screen broken"*) will not trigger auto-classification rules and require manual triage.
- Case sensitivity in keyword matches depends on the operator selected in Flow Designer.

13. Future Enhancements

- Incorporate **Predictive Intelligence / Machine Learning (PI)** to classify tickets using natural language processing (NLP) instead of static keyword matching.
- Add auto-assignment logic to directly routes tickets to specific support groups based on category.